

ABSTRACT

Stablecoins serve as a critical medium of exchange and store of value for trillions of dollars in daily transactions, yet they are not immune to losing their \$1.00 peg — triggering cascading losses and long-term erosion of issuer trust.

This project develops a machine learning early-warning system capable of predicting depeg events hours in advance, giving stablecoin issuers time to act before a crisis reaches open markets.

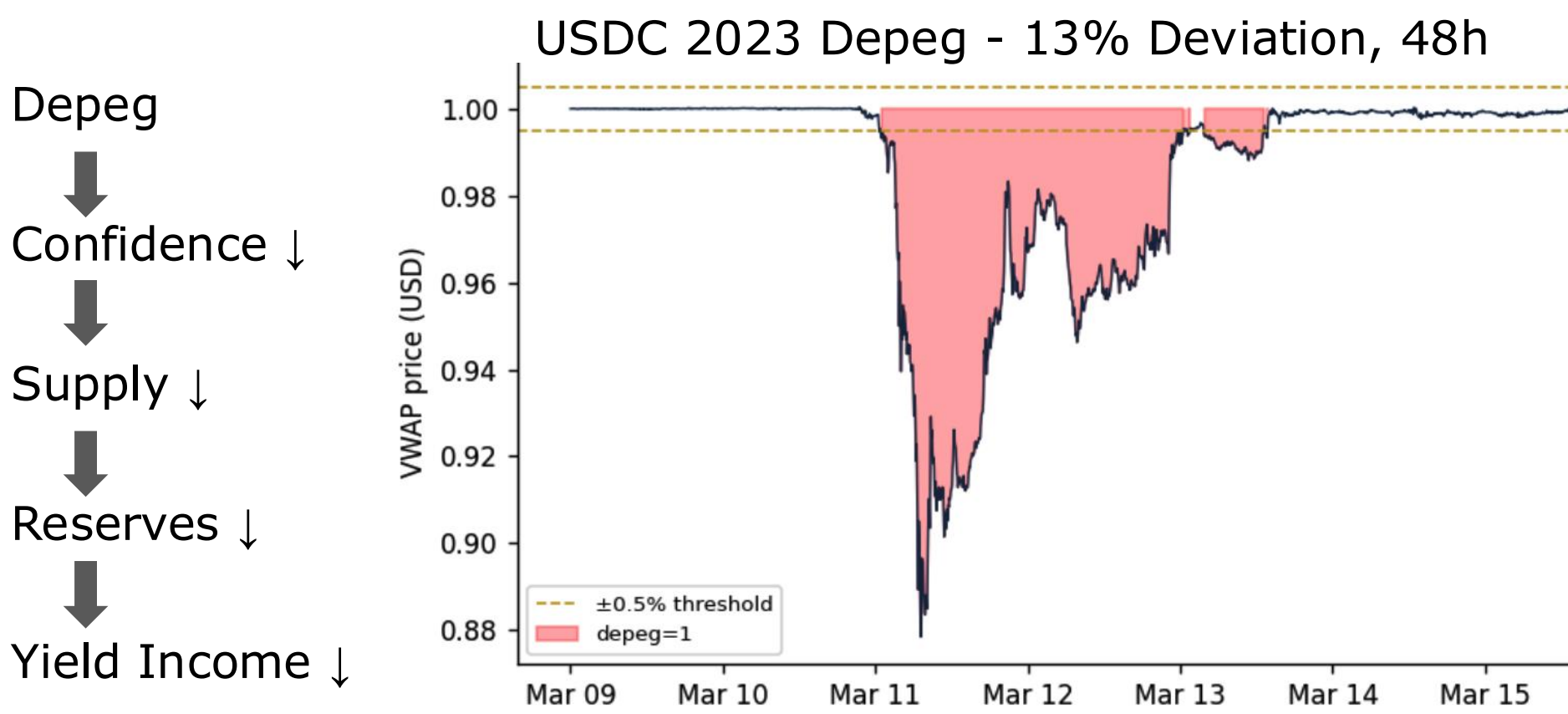
Drawing on price feeds, on-chain activity, DeFi liquidity signals, and macro indicators across seven stablecoins from 2017 to early 2026, we trained a predictive model on over 3.3 million observations — including historical failures such as TerraUSD — to identify early stress signals that precede a peg breakdown. The result transforms stablecoin monitoring from a reactive process into a forward-looking risk signal.

The result is a practical risk tool that transforms stablecoin monitoring from a reactive process into a forward-looking signal, ensuring risk managers are engaged at the right time to manage depeg risk.

INTRODUCTION

Stablecoins are a cornerstone of the modern crypto ecosystem, designed to maintain a fixed \$1.00 value and widely used for trading and payments.

Their value ultimately depends on trust in the peg. As the USDC depeg showed, once trust erodes — especially in severe events — panic-driven outflows can lead to substantial losses in future yield for issuers.



Circle’s estimated \$1.35 billion loss comes from roughly \$10 billion in outflows, which could have generated about 4.5% annual yield over a three-year period.

This project follows a four-step framework: problem framing and research, data sourcing and feature engineering, exploratory data analysis, and predictive modeling to detect and anticipate stablecoin depeg risk.

METHODS AND MATERIALS

We collected a dataset spanning seven stablecoins from 2017 to early 2026, (CoinAPI and Binance) drawing from four broad source categories: stablecoin price feeds, on-chain mint and burn activity, DeFi liquidity pool flows, and macro market indicators.

To being our project we had to first define our definition of a depeg. After a comprehensive literature review, we defined our definition of a depeg to be a price deviation to exceed 0.5% — equivalent to a price below \$0.995 or above \$1.005 — for a minimum of three consecutive 5-minute bars, representing a 15-minute sustained breach. We have severity metrics, that vary depending on threshold and duration, that we did not include but would be interesting for future research.

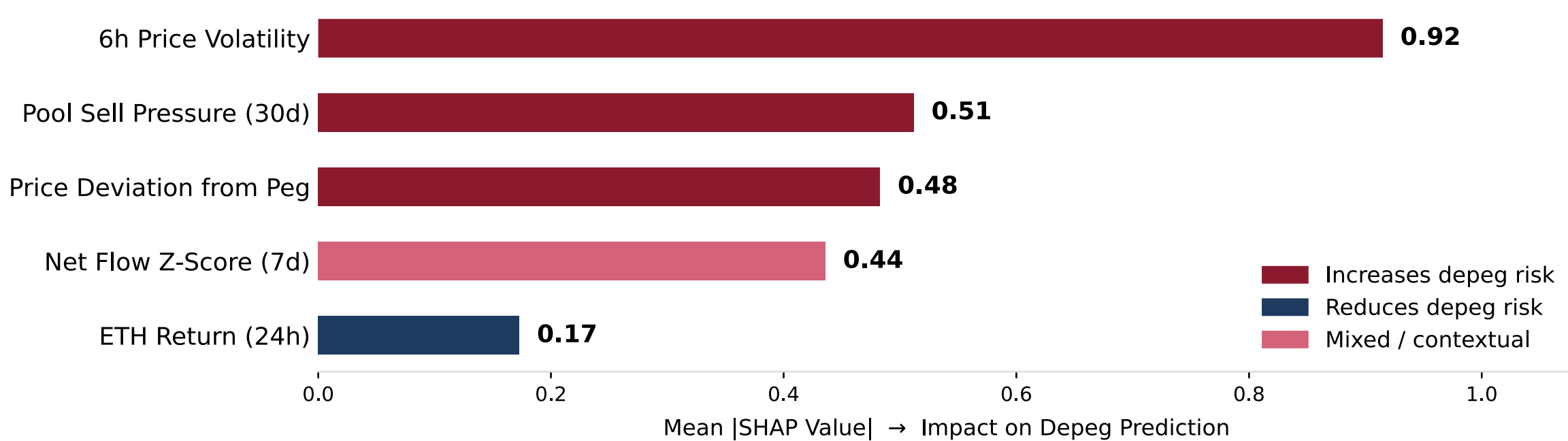
We then conducted descriptive analysis to identify potential features, examining price deviation patterns, on-chain flow dynamics, and market stress signals across historical depeg events. These insights informed the engineering of approximately 87 features per coin, refined through a systematic selection process, which we narrowed down to a selection of the most relevant features including: price deviation in basis points, one-hour and six-hour rolling volatility, six-hour momentum, a seven-day z-score, intrabar high-low range, trading volume surge and z-score measures, BTC and ETH 24-hour returns and volatility, VIX and DXY macro indicators, the Fear and Greed index, cross-coin price deviations for USDT, USDC, and DAI peers, maximum peer deviation, and the hour and day-of-week temporal features.

RESULTS

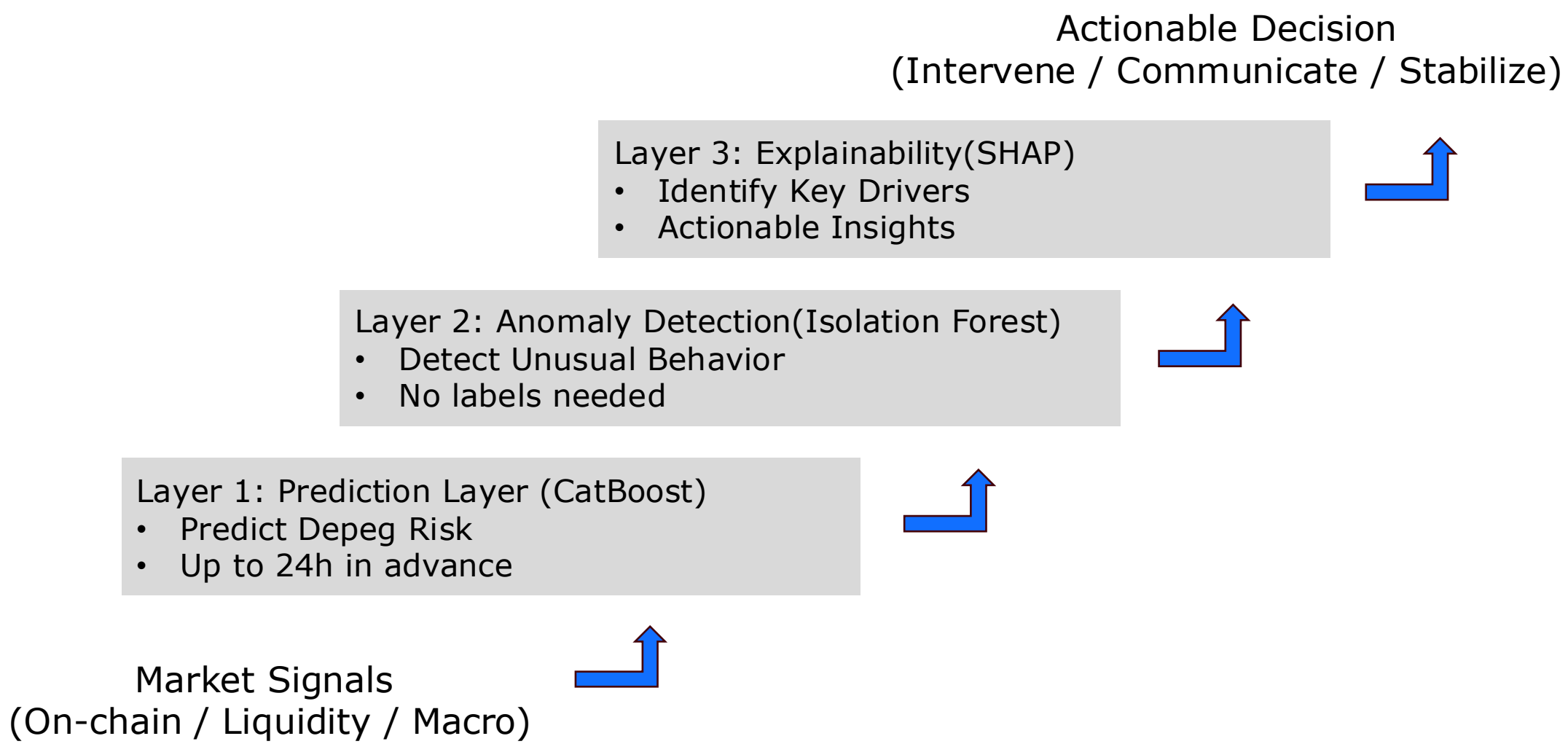
After evaluating nine machine learning algorithms, Catboost emerged as the strongest performer, supported by XGBoost and LightGBM as alternatives and an Isolation Forest added to catch abnormal market behavior outside any known pattern.

At best, our chosen model is 67 times more likely to flag a real depeg than random chance. Instead of scanning 1,000 price signals, it hands analysts a shortlist of 15 worth their attention. Tested against five real historical crisis, the model detected three events early. The UST collapse of May 2022 illustrate this perfectly. Alex, a fictional retail investor with \$50K in UST, lost 95% of his position without a warning. With our model, he exits at \$0.98 when the model triggers an alert and preserves everything. For the stablecoin issuer, the same alert could have prompted defensive action early enough to limit the LUNA dilution that accelerated the collapse.

Our model also reveals that 6-hour price volatility carries the strongest signal, nearly double the next feature. Positive ETH returns push the opposite way, confirming that broader market health suppresses depeg risk. Beyond individual rankings, the directional pattern across these signals offers a composite read on market health, where the tilt between risk-amplifying and stabilizing forces can reveal systemic stress before price reflects it.



A Three-Layer Defense System



DISCUSSION

Our analysis identifies two key data gaps that may limit the model’s ability to detect early warning signals. First, the lack of order book depth data prevents us from capturing liquidity thinning and spread widening, which often precede price deviations. Second, the absence of aggregated CEX flow data limits our ability to observe large-scale redemption or sell-side pressure driven by institutional behavior. Together, these missing signals represent critical components of pre-depeg market dynamics.

Incorporating these data sources could significantly enhance predictive performance, especially at short horizons. Both order book signals and CEX flows provide leading indicators of market stress, often emerging 5–30 minutes before visible price movements. By capturing these early signals, the model could improve its timeliness and accuracy, enabling more effective early intervention and ultimately reducing the economic impact of depeg events.

CONCLUSIONS

This study demonstrates that stablecoin depeg events are predictable potentially hours in advance.

Our results establish that on-chain institutional flows and DeFi liquidity signals are detectable leading indicators of open-market price dislocations — providing meaningful lead time for risk managers to act before a depeg becomes visible in spot prices.

Trust in the peg is the fundamental value proposition of stablecoins, without it stablecoins cannot function as a medium of exchange, collateral, or settlement layer in the payments ecosystem. Our modelling techniques can assist issuers to protect this trust, enabling them to grow market share and increase revenue while proving a valuable service to the digital finance ecosystem.